

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

1. Which of the following is NOT a reason it is important to have a well-designed database schema?

1 / 1 point

- ☐ Poorly designed database schemas can result in poor software performance
- ☐ Poorly designed schemas may not align well with all stakeholders' requirements of a piece of software
- ☒ Well designed schemas can eliminate the need to define the relationships between tables in your database
- ☐ Well designed schemas can be used by LLMs or programmers to quick prototype an application

✔ Correct
A schema, even if well defined, will still need to define the relationships between tables in your database,

2. Which of the following best describes the relationship between SQLite and SQLAlchemy?

1 / 1 point

- ☐ SQLite provides a lightweight database while SQLAlchemy provides cloud storage to host the database.
- ☐ SQLAlchemy provides a lightweight database while SQLite provides tools to query and interact with that database
- ☐ SQLite and SQLAlchemy are both databases. SQLite is used for small to medium-sized projects while SQLAlchemy is used for larger projects
- ☒ SQLite provides a lightweight database. SQLAlchemy provides tools to query and interact with that database

✔ Correct
Both products are used in managing a database, with SQLite actually providing for that database and SQLAlchemy providing tools to interact with it.

3. Which steps can an LLM help with while designing the schema for your database? Select all that apply.

1 / 1 point

- ☒ Translating high level business requirements into a preliminary schema

✔ Correct
LLMs are able to produce good draft database schemas when provided with a description of the requirements it needs to meet.

- ☒ Propose attributes and keys within each table of the database

✔ Correct
LLMs are capable of translating the attributes and relationships of entities in your database into features of your database schema like database attributes and keys

- ☒ Identify which keys can be used to match records across tables

✔ Correct
LLMs are able to generate code that manages these relationships within your data

- ☒ Write code to implement a prototype database schema

✔ Correct
LLMs are capable of writing code to implement a schema they are either provided or that they generate themselves.

4. Which CRUD operation is implemented by the following code?

1 / 1 point

```
1 def add_user(name, email):
2     new_user = User(name=name, email=email)
3     session.add(new_user)
4     session.commit()
```

- ☒ Create
- ☐ Read
- ☐ Delete
- ☐ Update

✔ Correct
This code is adding a new user record to the database and therefore is a "Create" operation.

5. In the videos the initial set of CRUD operations generated by the LLM were susceptible to SQL injection attacks. Which of the following approaches could be used to address these security risks?

1 / 1 point

- ☒ Give feedback to the LLM about potential security flaws you identify in code it initially generated

✔ Correct
Giving feedback to an LLM is a useful way to have it update code it initially generated with a new focus.

- ☐ Have the LLM regenerate its response using the same prompt to improve the odds of generating more secure code

- ☒ Assign the role of a security expert to the LLM before initially prompting it to increase the likelihood of using more secure methods

✔ Correct
Assigning a role to an LLM like this is a useful tool to initially generate code that aligns with your project's needs.

- ☒ Ask the LLM to review existing code from the standpoint of a security expert to identify potential issues not identified when the code was first generated

✔ Correct
Asking for expert review of code you already have, whether you wrote it or an LLM, is a good way to identify some issues and generate code closer to the final desired behavior.